

# Data Visualization in Business and Science

Prof. Dr. Do Phuc  
University of Information Technology  
Viet Nam National University Ho Chi Minh City



## Short Bio

Professor Dr. Do Phuc is currently working at the University of Information Technology, VNU-HCM. His research interests include bioinformatics, text processing, artificial intelligence, machine learning, deep learning, natural language processing, knowledge graphs, question answering systems, information verification, and big data processing.

Professor Do Phuc has participated in many research projects in various fields, including bioinformatics, text processing, natural language processing, text understanding systems, and reasoning over knowledge graphs. In the era of AI education, he has published more than 60 scientific papers in reputable international and national journals.

In addition, he is the author of 8 reference books and textbooks on computer science, programming techniques, social network analysis, big data analytics, data mining, knowledge graphs, and their applications.

Email: [phucdo@uit.edu.vn](mailto:phucdo@uit.edu.vn)



## Visualization & Information Visualization:

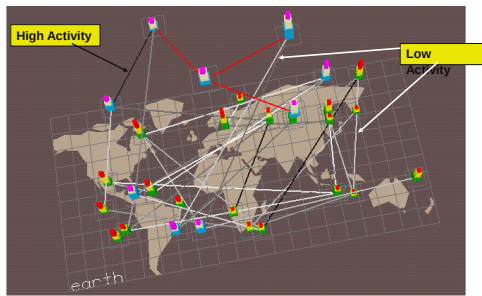
- Visualization:
  - “The use of computer-supported, interactive, visual representations of data to amplify cognition.”
  - Goal: discovery, decision making, explanation
- Information Visualization:
  - “The use of **computer-supported?**, interactive, visual representations of abstract data to amplify cognition.”

## Outline

- Information Visualization
- Graph visualization
- Data Visualization with Python
- Data Visualization of Google Scholar data
- Data Visualization of General Scientific Report
- Conclusion and future works

## Data Visualization in Business and Science Examples

## Global Private Network Activity

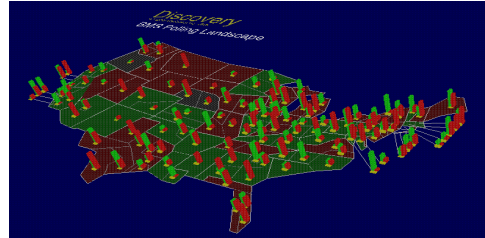


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## Telephone Polling Results



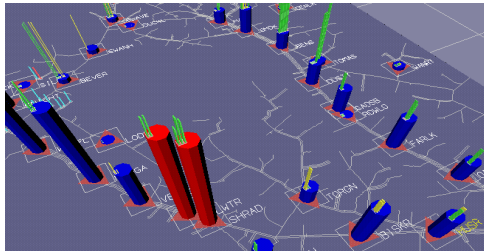
**Note:** On the “live” map, clicking on an area allows the user to drill down and see results for smaller areas.

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## Natural Gas Pipeline Analysis



**Note: Height shows total flow through compressor stations.**

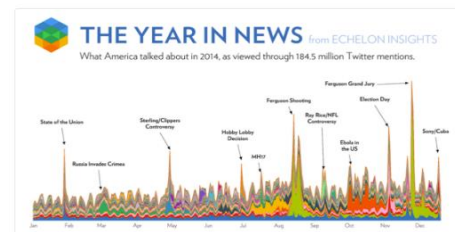
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## 12) The Year in News

The best data visualizations communicate information and do so in an intuitive and beautiful way. Echelon Insights nailed it with this piece, which visualizes the most talked-about new stories of 2014 on Twitter. What do 184.5 million tweets look like? Rad spin art.



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## An “Enlivened” Risk Analysis Report



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**Akami Real-time Web Monitor** shows a map of the world with real-time information about internet traffic, latency and current network attacks. The maps are color-coded and easy to read, but only give very generalized information.



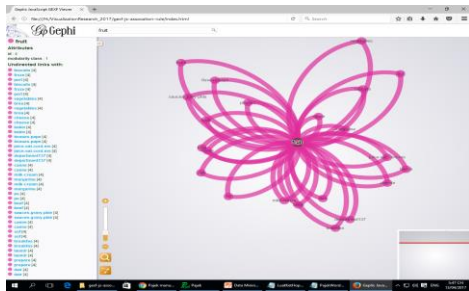
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## Graph Visualization by Gephi on the Web

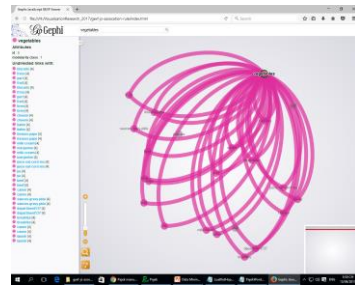


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## Discover the items cooccurring with vegetables

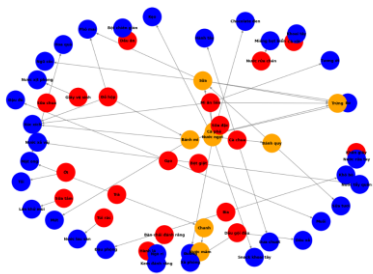


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## Graphs of AR



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## Using cube to visualize AR

- Each rule is a cube
- Click on Cube to visualize the association rule
- The height of Cube: the confidence of AR.
- Easy to discover the important rules based on the height of cube.

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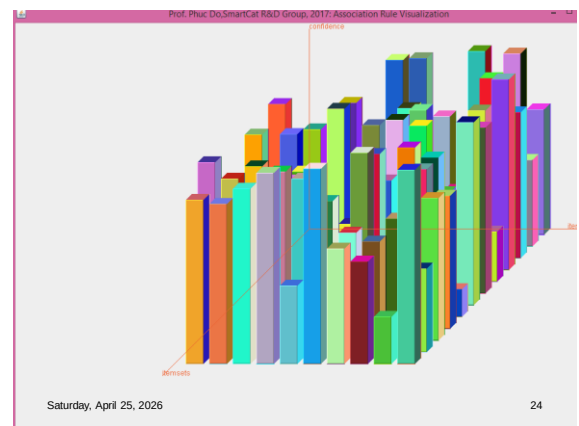
## Interactive mode in Graph visualization

- Very easy to capture the important association rules
- Click on node to discover the co-occurrence items
- Move around to discover the association rules

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## Visualization of Relations

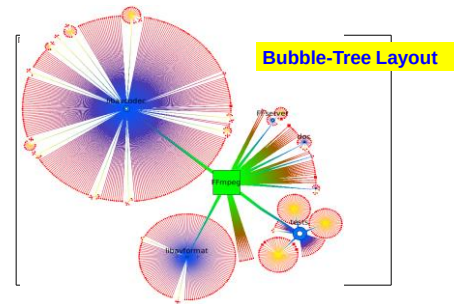
- Tree Visualization
- Graph Visualization

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## Tree Visualization



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## Tree Visualization

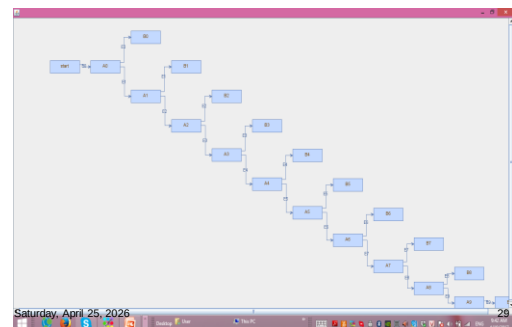
- Treemap method: visualize the tree structure that use virtually every pixel of the display space to convey information
- Every subtree is represented by a rectangle, that is partitioned into smaller rectangles with correspond to its children.
- The position of the slicing lines determines the relative sizes of the child rectangles.
- For every child, repeat the slicing direction from vertical to horizontal or conversely

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## Tree Visualization

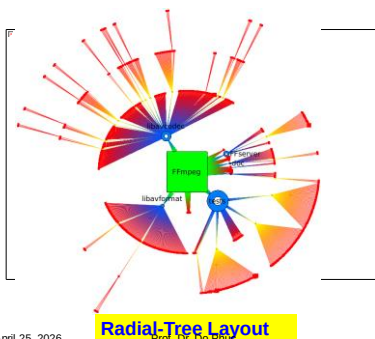


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## Tree Visualization

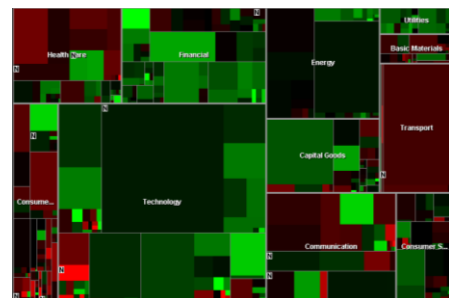


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## Tree Visualization



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## Graph Visualization

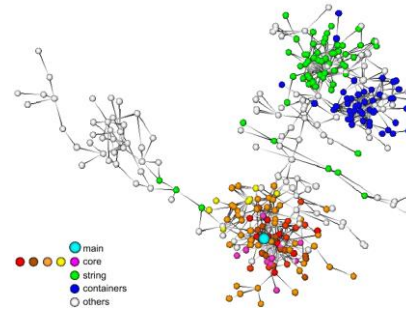
- Graphs are the most general type of relational data
- Similar to a tree, it consists of nodes and edges
- Different from a tree, a child node may have multiple parent nodes
- A graph contains loops, or multiple paths between two nodes in the graph.

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## Graph Visualization

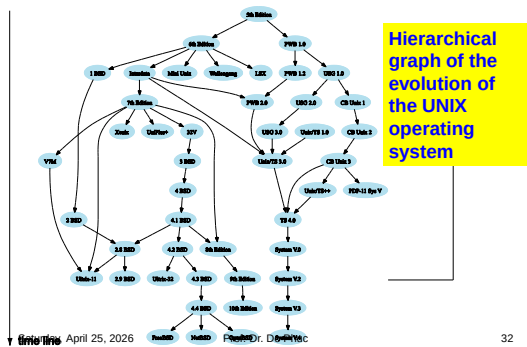


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Call Graph using a Force Directed Layout

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## Graph Visualization



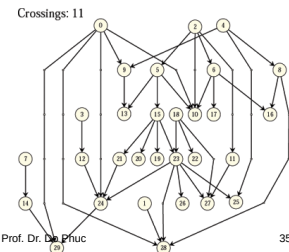
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## Layout of Directed Graphs

- Layering (<http://www.cs.usydney.edu.au/staff/NikolaNikolov/#phd>)

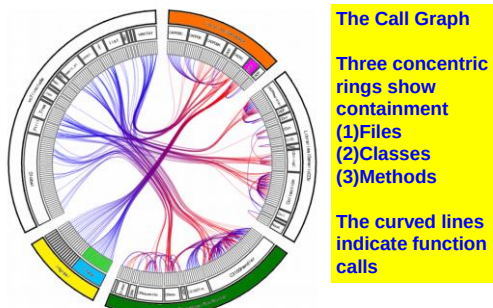


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## Graph Visualization



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## Spring Layout

- Force directed
- Nodes are modeled as physical bodies that are connected through springs (edges)
- High time complexity:  $> O(N^3)$
- Not predictable

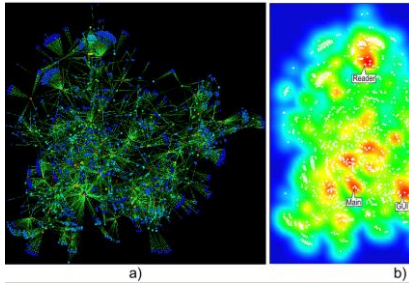
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## Graph Visualization



**Force-Directed Layout**

**Splatting;  
Dense Representation**

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## Co-authorship Network

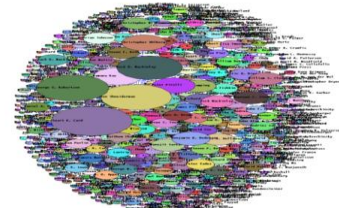


Figure 5. Concept map for co-authorship in the InfoVis dataset. Node size represents the number of citations for each author, and the strength of the springs binding two nodes is proportional to the amount of co-authored papers.

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## Text Visualization



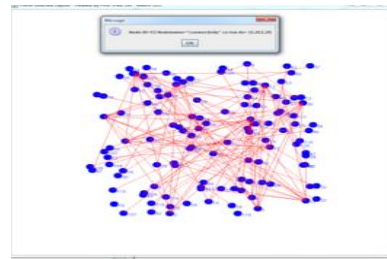
Visualize program code: colors denote the age (red-new, blue-old), and the pop-out window for details

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## Graph Visualization and Interaction



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## Graph Visualization Concept Map- Word Co-occurrence

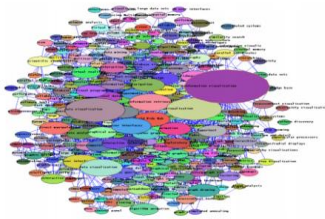


Figure 4. Concept map for the keywords in the InfoVis dataset. Node size represents the number of occurrences of a specific term, and links between nodes represent keywords that occur together.

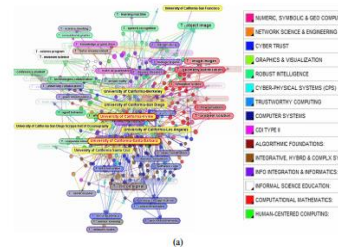
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## TopicNets: Visual Analysis of Large Text Corpora with Topic Modeling

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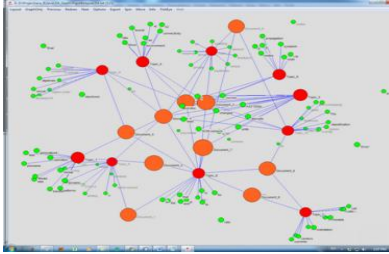
(a)

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## Topic Model LDA visualization



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## Chart Choice – Fearsome Foursome



**BARPLOT**  
Represents the value of entities using bar of various length.



**HISTOGRAM**  
An accurate graphical representation of the distribution of numeric data.



**SCATTER PLOT**  
Show the relationship between 2 numeric variables.

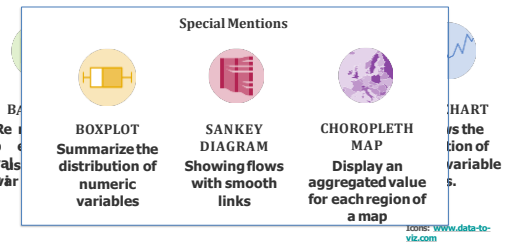


**LINE CHART**  
Shows the evolution of numeric variables.

Icons: [www.data-to-viz.com](http://www.data-to-viz.com)

## Chart Choice – Fearsome Foursome

## Data Visualization with Python

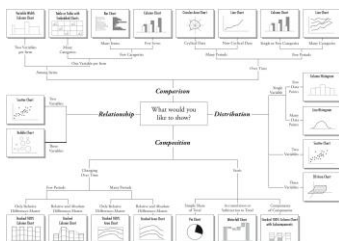


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## Chart Choice



Source: [www.extremerepresentation.com](http://www.extremerepresentation.com)

## Data Visualisation in Python



### Python Visualisation

- Lots of choice of libraries
- Many tools, with varied APIs & outputs
- Best to conquer and become familiar with one / two





## Matplotlib



Grand daddy of Python Plotting  
Low level plotting library  
with Matlab-like API

+ Very flexible, complete control  
- Verbose plots, aesthetically  
lacking, sometimes difficult  
with Pandas

...need to know enough to  
debug...



## Sample Data



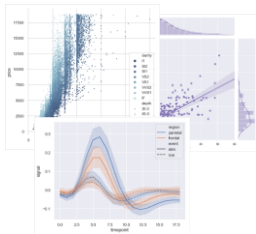
EdgeTier relevant sample dataset on chat system  
performance.  
Agents answering customer chats from different  
websites and languages – 5477 chats over 100  
weeks

```
In [7]: data.head()
```

```
Out [7]:
```

|   | user_id   | chat_id | language | number_messages | handling_time | start_time                 | website   | website_summary |
|---|-----------|---------|----------|-----------------|---------------|----------------------------|-----------|-----------------|
| 0 | User 1495 | 8347    | English  | 7               | 187.0         | 2019-05-30 11:59:49.422292 | Website B | Website B       |
| 1 | User 1495 | 8348    | English  | 7               | 258.0         | 2019-05-30 11:25:15.111164 | Website B | Website B       |
| 2 | User 1495 | 8349    | English  | 8               | 529.0         | 2019-05-30 10:40:38.253353 | Website B | Website B       |
| 3 | User 1495 | 8350    | English  | 14              | 840.0         | 2019-05-30 12:08:16.612382 | Website B | Website B       |
| 4 | User 1495 | 8351    | English  | 11              | 1283.0        | 2019-05-30 11:15:28.383306 | Website B | Website B       |

## Pandas / Seaborn / Altair



Higher level plotting

Pandas – Visualisation API built  
into DataFrame & Series objects,  
interface to Matplotlib.

Seaborn – extends and provides  
high-level API on Matplotlib  
with improved styling.

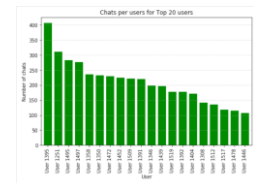
Altair – Built on “Vega-Lite”  
visualisation grammar. Allows  
some interactive plots in  
Jupyter Notebooks.

## The Bar Plot - Pandas



```
chats_per_user[0:20].plot(
    x='user_id', y='number_chats',
    kind='bar', legend=False, color='green',
    width=0.8
)
plt.ylabel('Number of chats')
plt.xlabel('User')
plt.title('Chats per users for Top 20 users')
plt.gca().yaxis.grid(linestyle=':')
```

Plot output is Matplotlib – same  
manipulation. Slightly simpler API /  
data access.



## Basic Notebook Setup



Imports on Matplotlib  
Top of notebook –  
inline vs notebook  
style.

Theme also can  
be chosen here

```
# Pandas used for data analysis and management
import pandas as pd
# Numpy library for array and numerical functions
import numpy as np
# Matplotlib pyplot provides plotting API
import matplotlib as mpl
from matplotlib import pyplot as plt
# "Standard" to load seaborn as "sns"
import seaborn as sns
# Altair is a visualisation library based on Vega
import altair as alt
alt.renderers.enable('notebook')

# For output plots inline in notebooks:
matplotlib inline
# For interactive plot controls on Matplotlib outputs:
%matplotlib notebook

# Set the default figure size for all inline plots
# (note: needs to be AFTER the matplotlib magic)
plt.rcParams['figure.figsize'] = [8, 5]
```

## The Bar Plot - Seaborn

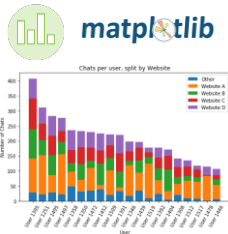


```
sns.barplot(x='user_id', y='number_chats',
            color='salmon', data=chats_per_user[0:20])
plt.xticks(rotation=60)
plt.ylabel('Number of chats')
plt.xlabel('User')
plt.title('Chats per users for Top 20 users')
plt.gca().yaxis.grid(linestyle=':')
```

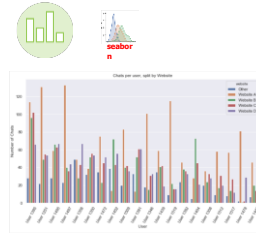
Simpler data access again.  
Same Matplotlib formatting  
functions



## Stacked Bar - Matplotlib



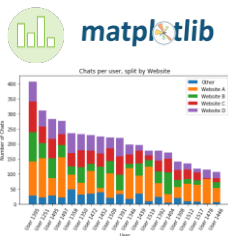
## Stacked Bar - Seaborn



```
sns.set()
sns.barplot(x='user_id',
            y='number_chats',
            hue='website',
            data=chats_per_user)
plt.xlabel('User')
plt.ylabel('Number of Chats')
plt.title('Chats per user, split by Website')
plt.xticks(rotation=45)
```

Elegant API, simple code structure, but ...  
...embarrassingly...  
no stacked-bar chart support!

## Stacked Bar - Matplotlib

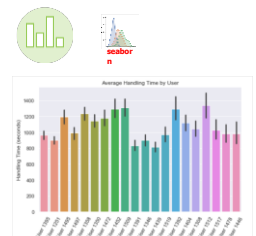


```
x_positoin = list(range(20))
# For matplotlib, we manually track where bars start/finish
bars_bottom = np.zeros(20)

for website_string in chats_per_user['website'].unique():
    # Draw a set of bars for this website
    bar_height = chats_per_user[
        chats_per_user['website'] == website_string]['number_chats']
    plt.bar(
        x=x_positoin,
        height=bar_height,
        bottom=bars_bottom
    )
    # Remember where the bottom of these bars are
    bars_bottom = bars_bottom + bar_height.values

# Unique preserves order - so it's safe here.
plt.xticks(x_positoin, chats_per_user.user_id.unique(), rotation=45)
plt.xlabel('User')
plt.ylabel('Number of Chats')
plt.title('Chats per user, split by Website')
plt.legend(chats_per_user['website'].unique())
```

## Seaborn - Estimators

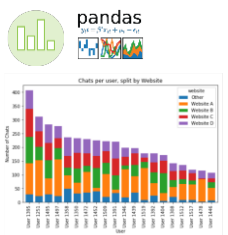


```
sns.barplot(
    x='user_id', y='handling_time', estimator=sns.mean,
    data=chats_per_user,
    order=data['user_id'].value_counts().index.tolist()[0:20])
plt.xlabel('User')
plt.ylabel('Handling Time (seconds)')
plt.title('Average Handling Time by User')
plt.xticks(rotation=45)
```

Calculations done as part of plotting – no previous data manipulations.

Separation of data and visualisation code.

## Stacked Bar - Pandas



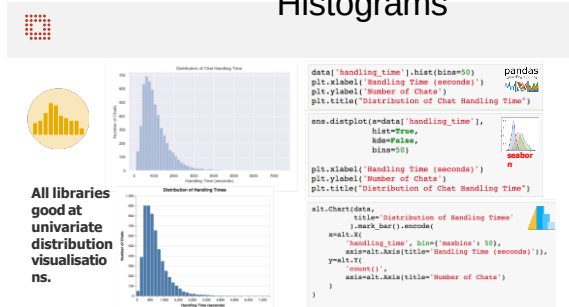
```
# Stacked bar in Pandas requires a data transformation
plot_data = chats_per_user.pivot(index='user_id',
                                columns='website',
                                values='number_chats')
# Wide format - sort by total chats by user.
plot_data = plot_data.sort_values('sum(axis=1)', ascending=False)
plot_data.head()
```

| user_id   | Other | Website A | Website B | Website C | Website D |
|-----------|-------|-----------|-----------|-----------|-----------|
| User 1000 | 25    | 400       | 100       | 100       | 100       |
| User 1001 | 100   | 400       | 100       | 100       | 100       |
| User 1002 | 100   | 400       | 100       | 100       | 100       |
| User 1003 | 100   | 400       | 100       | 100       | 100       |
| User 1004 | 100   | 400       | 100       | 100       | 100       |
| User 1005 | 100   | 400       | 100       | 100       | 100       |
| User 1006 | 100   | 400       | 100       | 100       | 100       |
| User 1007 | 100   | 400       | 100       | 100       | 100       |
| User 1008 | 100   | 400       | 100       | 100       | 100       |
| User 1009 | 100   | 400       | 100       | 100       | 100       |
| User 1010 | 100   | 400       | 100       | 100       | 100       |

```
plot_data.plot(kind='bar', stacked=True, width=0.8)
plt.xlabel('User')
plt.ylabel('Number of Chats')
plt.title('Chats per user, split by Website')
```

Plotting code is simple, but data manipulation required.

## Histograms



All libraries good at univariate distribution visualisation ns.

```
data['handling_time'].hist(bins=50)
plt.xlabel('Handling Time (seconds)')
plt.ylabel('Number of Chats')
plt.title('Distribution of Chat Handling Times')

sns.distplot(data['handling_time'],
             hist=True,
             kde=False,
             bins=50)
plt.xlabel('Handling Time (seconds)')
plt.ylabel('Number of Chats')
plt.title('Distribution of Chat Handling Times')

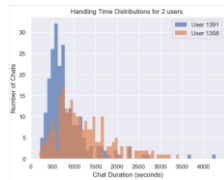
alt.Chart(data,
           title='Distribution of Handling Times')
    .mark_bar()
    .encode(
        x=alt.X(
            'handling_time', bin='auto',
            axis=alt.Axis(title='Handling Time (seconds)'),
            count=True
        ),
        y=alt.Y(
            'count()'
        )
    )
```

## Histograms



Layering / comparison achieved unfortunately by building up the histograms in place.

```
data.loc[data['user_id'] == 'User 1391', 'handling_time'].hist(
    bins=50, alpha=0.7, label='User 1391')
data.loc[data['user_id'] == 'User 1358', 'handling_time'].hist(
    bins=50, alpha=0.7, label='User 1358')
plt.legend()
plt.title('Handling Time Distributions for 2 users.')
plt.ylabel('Number of Chats')
plt.xlabel('Chat Duration (seconds)')
```



## Line Plots

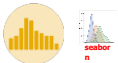


Pandas: Needs data manipulation, simple

```
# Pandas again, for multiple lines requires data manipulation
pandas_plot = plot_data.pivot(
    index='date',
    columns='language',
    values='number_chats'
).fillna(0)
# Actual plotting command is simple
pandas_plot.plot(kind='line', marker='o', linestyle='r')
# Formatting arguments:
plt.xlabel('Date')
plt.xticks(rotation=90)
plt.ylabel('Number Chats')
plt.title('Chats per day per language')
```

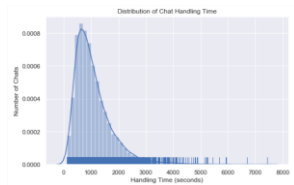


## Histograms - Seaborn



```
sns.distplot(a=data['handling_time'],
             hist=True,
             kde=True,
             rug=True,
             bins=50)
plt.xlabel('Handling Time (seconds)')
plt.ylabel('Number of Chats')
plt.title('Distribution of Chat Handling Time')
```

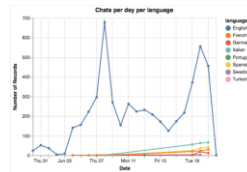
Some really nice options for impressive and informative hints on Seaborn graphs.



## Line Plots



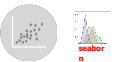
Seaborn/Altair: Operate directly on raw data



```
# Seaborn can act directly on the raw data!
sns.lineplot(
    x='date',
    y='chat_id',
    hue='language',
    data=data,
    estimator='mean',
    naxt='o'
)
# Formatting of plot output
plt.xlabel('Date')
plt.xticks(rotation=90)
plt.ylabel('Number Chats')
plt.title('Chats per day per language')

# Altair works directly on the raw data
alt.Chart(
    data,
    title='Chats per day per language'
).mark_line(point=True).encode(
    alt.X('date', title='Date'),
    y='count(chat_id)',
    color='language'
)
```

## Scatter Plots - Seaborn



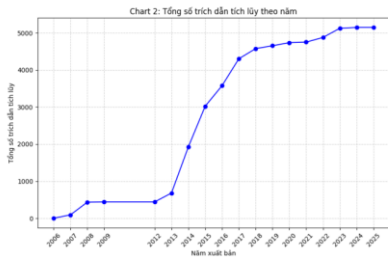
Seaborn / Altair: Better higher level representation, and better for multi-category scatters.

```
sns.lmplot(
    x='handling_time',
    y='number_messages',
    hue='vehicle_category',
    data=data,
    fit_reg=False, # DO NOT fit a regression line by default
    scatter_kws={'s': 10}, # Size of points on figure
)
plt.title('Handling Time vs Number of Messages')
plt.xlabel('Handling Time (seconds)')
plt.ylabel('Number of Messages')
```



## Data Visualization with data from Google Scholar

Biểu đồ 2:

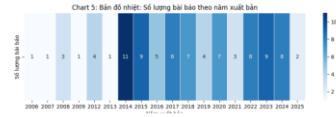


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Biểu đồ 5:

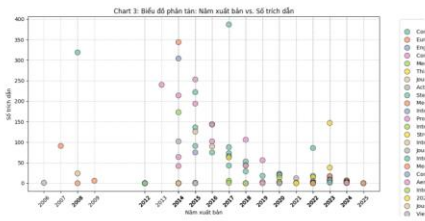


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Biểu đồ 3:

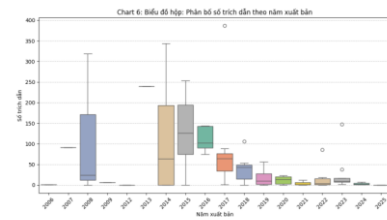


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Biểu đồ 6:

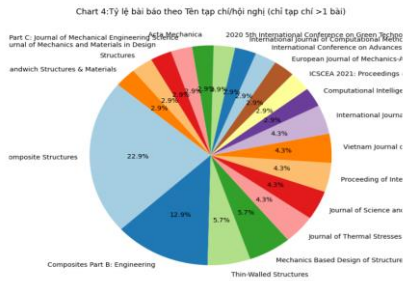


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Biểu đồ 4:

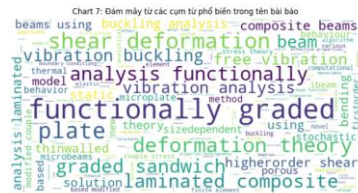


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Biểu đồ 7:

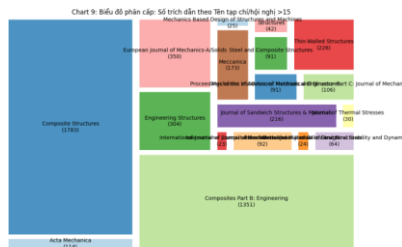


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**Biểu đồ 9:**



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CỘNG HÒA XÃ HỘI CHỦ NGHĨA VIỆT NAM  
Độc lập - Tự do - Hạnh phúc

## BÁO CÁO KHOA HỌC TỔNG QUAN

## I. THÔNG TIN CHUNG

1. Họ và tên ứng viên: .....
2. Cơ quan công tác hiện nay (Bộ môn, khoa, phòng, ban; trường, viện; thuộc Bộ):  
Khoa Công Nghệ Thông Tin; Trường Đại học Công nghệ TP. HCM
3. Chuyên ngành được đào tạo: Khoa học Máy tính
4. Môn học, chuyên ngành đang tham gia đào tạo: Khoa học Máy tính.

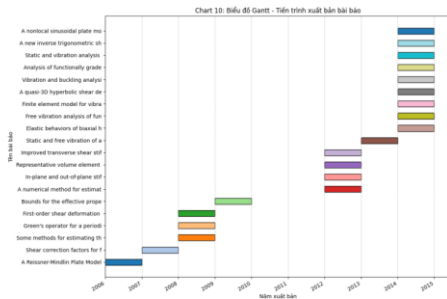
## II. NỘI DUNG

## 1. Mở đầu

Với sự tiến bộ của khoa học và kỹ thuật, bao gồm khả năng tính toán và khả năng thu thập dữ liệu lớn, nghiên cứu về máy học đã và mang lại nhiều ứng dụng hữu ích trong đời sống xã hội từ gia đình, đến các lĩnh vực an ninh, giáo dục, y tế, v.v. Do đó, *máy học và ứng dụng trong phân tích dữ liệu y sinh và an toàn người bệnh* được chọn là hai hướng nghiên cứu chính của tôi.

Cả hai hướng nghiên cứu này được thực hiện từ khi tôi công tác tại trường Đại học Bách Khoa, Đại học Quốc gia Thành phố Hồ Chí Minh; trong thời gian làm tiến sĩ và sau tiến sĩ tại Đại học Quốc gia Chonnam (Chonnam National University), Hàn Quốc; và trong thời gian làm học giả (visiting scholar) tại Đại học Chicago (University of Chicago), Hoa Kỳ; và được

**Biểu đồ 10:**

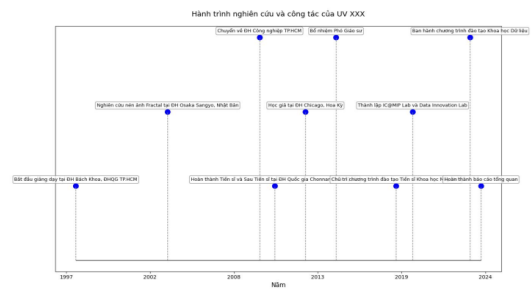


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**Biểu đồ 1:**



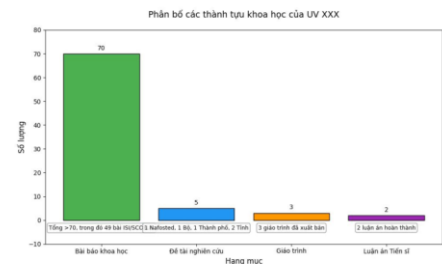
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## General Scientific report

**Biểu đồ 2:**



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Biểu đồ 3:

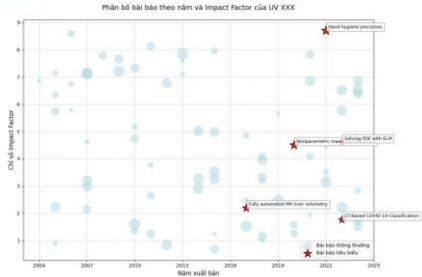


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Biểu đồ 6:

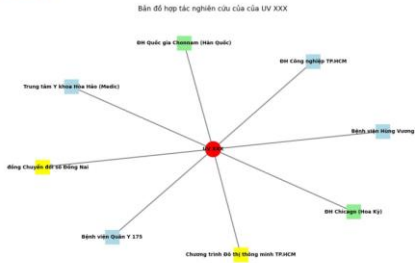


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Biểu đồ 4:

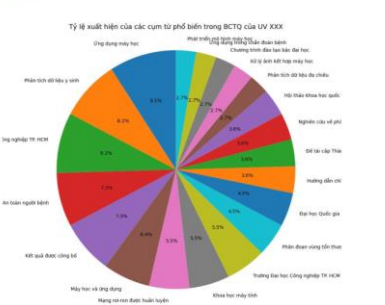


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Biểu đồ 7:

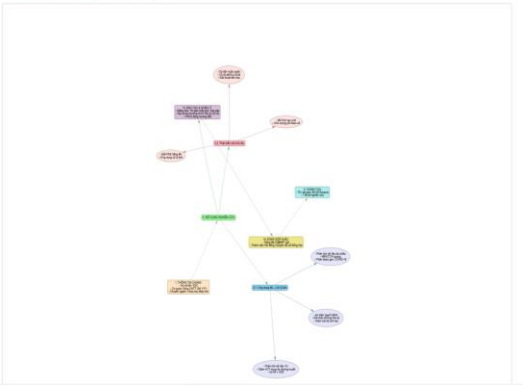


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Biểu đồ 8: MindMap từ BCTQ

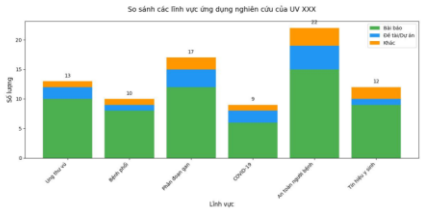


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Biểu đồ 5:



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graph neural representation learning classification molecular informatics such as molecular graph topological enhanced multi-view contrastive approach representation learning and classification paper in recent years node and edge multiview graph graph embedding graph neural network designed to molecular graph topological enhanced multi-view contrastive approach representation learning and classification paper

## Conclusion and future Works

- Data visualization bridges the gap between raw data and actionable insights.
- Effective visuals enhance understanding, reveal trends, and support decision-making.
- Choosing the right chart type and design principles is critical for clarity and impact.
- Interactive and real-time dashboards are becoming essential tools in data-driven environments

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**Mind map of Paper entitled 'White Blood Cell Segmentation using Kolmogorov-Arnold Network coupled with UNet'**

- Pre-processing
  - Image Acquisition
  - Image Pre-processing
  - Image Augmentation
- Dataset
  - Dataset Description
  - Dataset Pre-processing
  - Dataset Augmentation
- Model
  - Model Architecture
  - Model Training
  - Model Evaluation
- Evaluation
  - Evaluation Metrics
  - Evaluation Results
- Results
  - Results Discussion
  - Results Conclusion
- Conclusion
  - Conclusion Discussion
  - Conclusion Conclusion
- Future Work
  - Future Work Discussion
  - Future Work Conclusion
- References
  - References Discussion
  - References Conclusion

## Future works

- Explore AI-assisted visualization tools for automating insights.
- Integrate visualization into big data and streaming pipelines.
- Apply AR/VR for immersive data experiences.
- Emphasize ethical visualization — avoid misleading representations.

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## Conclusion and future Works

- Data visualization bridges the gap between raw data and actionable insights.
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- Interactive and real-time dashboards are becoming essential tools in data-driven environments

Thank you for your attention!

Please contact. Prof. Phuc Do  
Email: [phucdo@uit.edu.vn](mailto:phucdo@uit.edu.vn)

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